Year 2 Statistics and Mechanics

Chapter 8

Further kinematics

Practice Booklet

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8.1 Vectors in kinematics

Practice Book, printed page 68. In these questions, **i** and **j** are the unit vectors due east and north respectively.

1. A particle starts from the point with position vector $(4\mathbf{i} + 2\mathbf{j})$ m and moves with constant velocity $(\mathbf{i} - 3\mathbf{j})$ m s^{-1}. Work out the position of the particle after 5 seconds.

Hint

Use $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$ to find the position of the particle.

2. A particle starts from the point with position vector $(-6\mathbf{i} + \mathbf{j})$ m and moves with constant velocity $(2\mathbf{i} + \mathbf{j})$ m s^{-1}.

a. Work out the position of the particle after 2 seconds.

b. Find the time when the particle is due north of the origin.

Hint

When the particle is due north of the origin the **i** component of its displacement will be zero.

3. A particle passes a point P with velocity $(30\mathbf{i} + 20\mathbf{j})$ m s^{-1} and moves with constant acceleration $(10\mathbf{i} + 20\mathbf{j})$ m s^{-2}. Work out:

a. the displacement of the particle 2 seconds after it passes P

b. its distance from P at this time.

Hint

Use $\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$. Distance is the magnitude of the displacement vector.

4. A particle has velocity $(3\mathbf{i} - 2\mathbf{j})$ m s^{-1} and moves with constant acceleration $(3\mathbf{i} + \mathbf{j})$ m s^{-2}. Work out the speed of the particle and the bearing on which it is travelling after 3 seconds.

Hint

Use $\mathbf{v} = \mathbf{u} + \mathbf{a}t$. To work out the bearing draw a triangle showing the velocity vector and work out the angle clockwise from north to the velocity vector.

5. A runner runs on a horizontal playing field. At time $t = 0$, the runner is at a fixed point O and has a velocity of $(-8\mathbf{i} + 3\mathbf{j})$ m s^{-1}. After 8 seconds, the runner has a velocity of $(4\mathbf{i} - 5\mathbf{j})$ m s^{-1}. By modelling the runner as a particle moving with constant acceleration, work out the acceleration of the runner.

6. At time $t = 0$ a toy car has velocity $(a\mathbf{i} + b\mathbf{j})$ m s^{-1}. It accelerates at $(3\mathbf{i} + \mathbf{j})$ m s^{-2} for 5 seconds, after which time it has velocity $(4\mathbf{i} - 13\mathbf{j})$ m s^{-1}.

a. Work out the values of a and b.

b. Calculate the distance travelled by the car.

7. E An aeroplane starts from the point P with position vector $\begin{pmatrix}1540\\2500\end{pmatrix}$ m relative to a fixed origin. The aeroplane moves with constant velocity $\begin{pmatrix}-220\\-400\end{pmatrix}$ m s^{-1}. Find:

a. the position vector of the aeroplane after t seconds (1 mark)

b. the distance of the aeroplane from the origin after 2 seconds (2 marks)

c. the time when the aeroplane is due south of the origin. (2 marks)

8. E A toy train moves with constant acceleration. At time $t = 0$ seconds, the train has velocity $\begin{pmatrix}2\\-5\end{pmatrix}$ m s^{-1}, and at time $t = 5$ seconds, it has velocity $\begin{pmatrix}14\\5\end{pmatrix}$ m s^{-1}.

a. Find the initial speed of the train. (2 marks)

b. Calculate the acceleration of the train. (2 marks)

Given that the initial position of the toy train relative to a fixed origin is $(-\mathbf{i} + 8\mathbf{j})$ m,

c. find the position vector of the train after 6 seconds. (2 marks)

9. E/P At time $t = 0$, two walkers, A and B, have position vectors relative to a fixed origin given by $\begin{pmatrix}4\\-6\end{pmatrix}$ km and $\begin{pmatrix}18\\12\end{pmatrix}$ km respectively. A walks with velocity $\begin{pmatrix}1.5\\2\end{pmatrix}$ km h^{-1} and B walks with velocity $\begin{pmatrix}-2\\-2.5\end{pmatrix}$ km h^{-1}.

a. Show that the two walkers will meet. (3 marks)

b. Work out the position vector of the point where they meet. (2 marks)

10. E/P A particle P of mass 2 kg moves under the action of a constant force of $\mathbf{F}$ N. At time $t = 0$ seconds, the particle is at a point A and is travelling with velocity $(5\mathbf{i} + 10\mathbf{j})$ m s^{-1}. At time $t = 3$ seconds, the particle is at a point B and is travelling with velocity $(14\mathbf{i} - 2\mathbf{j})$ m s^{-1}.

a. Find $\mathbf{F}$, giving your answer in the form $a\mathbf{i} + b\mathbf{j}$. (5 marks)

At the instant that P passes point B, the force is removed.

b. Find the time that further elapses before the particle is directly due east of point A. (5 marks)

8.2 Vector methods with projectiles

Practice Book, printed page 69. In these questions, you may assume that $\mathbf{i}$ and $\mathbf{j}$ are unit vectors acting horizontally and vertically respectively, and that $g = 9.8$ m s^{-2}.

1. A particle is projected from a point on a horizontal plane with an initial velocity of $\begin{pmatrix}15\\25\end{pmatrix}$ m s^{-1}. The particle moves freely under gravity. Find:

a. the velocity of the particle 2 seconds after it is projected

b. its speed at this time.

Hint

Use $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ with $\mathbf{a} = -9.8\mathbf{j}$ m s^{-2}

2. At time $t = 0$, a ball is projected from a point P, which has position vector $\begin{pmatrix}6\\6\end{pmatrix}$ m relative to a fixed origin O. Immediately after projection, the ball has velocity $\begin{pmatrix}12\\19\end{pmatrix}$ m s^{-1}. The ball is modelled as a particle moving freely under gravity.

a. Find an expression for the displacement of the ball from P at time t, giving your answer in the form $\begin{pmatrix}x\\y\end{pmatrix}$ m, where x and y are functions of t.

b. Hence find an expression for the position vector of the ball relative to O.

Hint

Use $\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$ to find the displacement.

3. At time $t = 0$, an object is projected from a point A which has position vector $\begin{pmatrix}0\\15\end{pmatrix}$ m relative to a fixed origin O that lies on a horizontal plane. Immediately after projection, the object has velocity $\begin{pmatrix}23\\13\end{pmatrix}$ m s^{-1}. The object moves freely under gravity before landing at the point B.

a. Find an expression for the position vector of the object relative to O.

b. Work out:

i. the time taken for the object to reach B

ii. the distance OB.

Hint

When the object reaches B, the **j**-component of the position vector is zero.

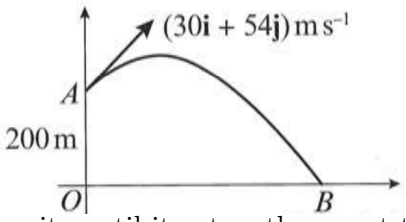
4. A particle is projected from a point on level ground with initial velocity $(8\mathbf{i} + 7\mathbf{j})$ m s^{-1}. Work out the greatest height reached by the particle.

5. E/P A particle is projected with velocity $\begin{pmatrix}a\\b\end{pmatrix}$ m s^{-1} from a point O that lies on a horizontal plane. The particle moves freely under gravity. The particle lands after 5 seconds at the point B where $OB = 120$ m. Work out:

a. the value of a and the value of b (3 marks)

b. the greatest height reached by the particle. (2 marks)

6. E A particle is projected with velocity $(30\mathbf{i} + 54\mathbf{j})$ m s^{-1} from a point A with position vector $200\mathbf{j}$ m relative to a fixed origin O, which is at sea level.



The particle moves freely under gravity until it enters the sea at the point B. Work out:

a. the time taken for the particle to enter the sea (4 marks)

b. the distance AB. (3 marks)

7. E/P A particle is projected with velocity $(13\mathbf{i} + 20\mathbf{j})$ m s^{-1} from a point O that lies on horizontal ground. The particle moves freely under gravity until it lands on a platform. The platform is at a height of 20 m above the ground. Work out:

a. the horizontal distance travelled by the particle (4 marks)

b. the speed with which the particle strikes the platform. (3 marks)

8. E/P A particle is projected with velocity $(p\mathbf{i} + q\mathbf{j})$ m s^{-1} from the origin O that lies on horizontal ground. The particle moves freely under gravity until it lands on the ground 6 seconds later with speed 13 m s^{-1} at an angle of α to the horizontal, where $\tan\alpha = \frac{12}{5}$. Work out the values of p and q. (5 marks)

8.3 Variable acceleration in one dimension

Practice Book, printed page 71.

1. A particle is moving in a straight line. At time t seconds, the displacement, s m, of the particle from the origin is given by $s = 2\sin \pi t + t^2$. Find:

a. the velocity of the particle when $t = 2$

b. the acceleration of the particle when $t = 5$.

Hint

Differentiate displacement to find velocity: $v = \dfrac{\mathrm{d}s}{\mathrm{d}t}$

Differentiate velocity to find acceleration: $a = \dfrac{\mathrm{d}v}{\mathrm{d}t}$

← **Year 1, Section 11.2**

2. A particle starts at rest at the origin, O, and moves in a straight line. At time t seconds, the acceleration, a m s^{-2}, of the particle is given by $a = \mathrm{e}^t + t^2$. Find:

a. an expression for the velocity of the particle

b. the displacement of the particle from the origin when $t = 3$.

Hint

To find the velocity, integrate the expression for the acceleration. Then use the initial condition that $v = 0$ when $t = 0$ to find the value of the constant of integration.

3. E/P A particle moves in a straight line such that at time t seconds, its displacement s m from a fixed point O is given by:

$$s = \begin{cases} 2t + 4^t - 4, & 0 \leq t \leq 2 \\ t(t+6), & t > 2 \end{cases}$$

Work out:

a. the velocity of the particle when $t = 1.5$ (3 marks)

b. the acceleration of the particle when $t = 5$. (2 marks)

4. E/P A particle is moving in a straight line. At time t seconds, its acceleration is given by

$$a = \frac{2}{t^2} - \cos 2\pi t \text{ m s}^{-2}, \ t \geq 1$$

Given that when $t = 1$, $v = 0$ and $s = 0$, find an expression for the displacement of the particle from the origin at time t seconds. (8 marks)

5. E/P A particle moves in a straight line, passing point O at $t = 0$ seconds. At time t seconds, $t \geq 0$, its velocity is given by $v = kt\cos t$ m s^{-1}. Given that when $t = \pi$ the particle has acceleration 0.5 m s^{-2}, work out:

a. the value of k (3 marks)

b. an expression for the displacement of the particle from O (5 marks)

c. the displacement of the particle from O when $t = \pi$. (1 mark)

6. E/P A particle P of mass 2 kg moves in a straight line under the action of a single variable force of magnitude F N. At time t seconds, where $t \geq 0$, the displacement, s m, of P from a fixed point O is given by $s = 1 - \mathrm{e}^{-kt}$. Given that at $t = 0$ the velocity of the particle is 2 m s^{-1}, work out:

a. the value of k (3 marks)

b. the value of F when $t = 0.5$. (5 marks)

7. E/P A particle is moving in a straight line. At time t seconds, the displacement, s m, of the particle from the origin is given by $s = \sqrt{t^2 + 15}$, where $0 \leq t \leq 7$.

a. Find an expression for the velocity of the particle. (3 marks)

b. Show that the velocity of the particle is increasing on the interval $0 \leq t \leq 7$, and hence state the maximum speed of the particle. (5 marks)

8.4 Differentiating vectors

Practice Book, printed page 72. In these questions, you may assume that **i** and **j** are the unit vectors due east and north respectively.

1. At time t seconds, the position vector $\mathbf{r}$ m of a particle relative to a fixed origin is given by

$$\mathbf{r} = 3t^2\mathbf{i} - 5\sqrt{t}\mathbf{j},\ t \geq 0$$

a. Work out the velocity of the particle when $t = 9$.

b. Calculate the speed and direction in which the particle is moving when $t = 1$.

Hint

Use $\mathbf{v} = \dfrac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$. You need to differentiate the **i**-component and the **j**-component separately.

2. At time t seconds the velocity $\mathbf{v}$ m s^{-1} of a particle P is given by

$$\mathbf{v} = \cos t\mathbf{i} - 5t\mathbf{j},\ t \geq 0$$

Work out:

a. the acceleration of P when $t = \pi$

b. the direction in which P is accelerating at $t = \dfrac{\pi}{2}$.

Hint

Use $\mathbf{a} = \dfrac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}$

3. At time t seconds a particle P has position vector $\mathbf{r}$ m relative to a fixed origin, where

$$\mathbf{r} = t(t+1)\mathbf{i} + (t^3 - 1)\mathbf{j},\ t \geq 0$$

Work out:

a. the distance of P from the origin when $t = 2$

b. the acceleration of P when $t = 4$, giving your answer in the form $(a\mathbf{i} + b\mathbf{j})$ m s^{-2}.

Hint

The distance is the magnitude of the displacement vector.

4. E/P At time t seconds a particle P has position vector $\mathbf{r}$ m relative to a fixed origin, where

$$\mathbf{r} = \mathrm{e}^{3t}\mathbf{i} + \sin 2\pi t\mathbf{j},\ t \geq 0$$

Work out the acceleration of the particle when $t = 2$. (6 marks)

5. E/P A particle P is moving in a plane. At time t seconds, it has a position vector relative to a fixed origin O of $\mathbf{r} = \left(4t\mathbf{i} - \dfrac{3}{t+1}\mathbf{j}\right)$ m, $t \geq 0$. Work out:

a. the time when the particle is directly south-east of O (3 marks)

b. the speed of the particle at this time. (3 marks)

6. E/P A particle P is initially at a fixed origin O. P moves such that, at time t seconds, it has a position $\mathbf{r}$ m relative to O, where $\mathbf{r} = (2 - 3t^2)\mathbf{i} - t(t^3 - 4)\mathbf{j}$, $t \geq 0$. Work out:

a. the initial speed of P (3 marks)

b. the time when P is moving parallel to **i** (2 marks)

c. the displacement of P when it is moving parallel to **i**. (2 marks)

7. E/P At time t seconds a particle P has displacement $\mathbf{r}$ m relative to a fixed origin, where

$$\mathbf{r} = 3e^{2t(t+1)}\mathbf{i} - e^{t}\mathbf{j},\ t \geq 0$$

a. Find the speed of P when $t = 0.2$. (4 marks)

b. Work out the acceleration of P giving your answer in vector form. (6 marks)

8. E/P A particle P moves under the action of a single force $\mathbf{F}$ N. At time t seconds, where $t \geq 0$, the position vector of the particle $\mathbf{r}$ m, relative to a fixed origin, is given by $\mathbf{r} = 5t(t+1)\mathbf{i} + (3 - t^2)\mathbf{j}$.

a. Show that P moves with constant acceleration. (5 marks)

b. Given that the mass of P is 5 kg, find the magnitude of $\mathbf{F}$. (3 marks)

8.5 Integrating vectors

Practice Book, printed page 73. In these questions, you may assume that $\mathbf{i}$ and $\mathbf{j}$ are the unit vectors due east and north respectively.

1. A particle is moving in a plane so that at time t seconds, its acceleration $\mathbf{a}$ m s^{-2} is given by

$$\mathbf{a} = 5t\mathbf{i} + 2\mathbf{j},\ t \geq 0$$

When $t = 2$, the velocity of the particle is $(8\mathbf{i} + 2\mathbf{j})$ m s^{-1}.

a. Find an expression for the velocity of the particle at time t, giving your answer in the form $(f(t)\mathbf{i} + g(t)\mathbf{j})$ m s^{-1}.

b. Calculate the speed of the particle at $t = 3$.

Hint

$\mathbf{v} = \int \mathbf{a}\,dt$
Integrate and include a vector constant of integration, $\mathbf{c}$. Then use the condition given in the question to find $\mathbf{c}$.

2. A particle is moving in a plane such that, at time t seconds, it has velocity $(3\mathbf{i} - 2t^2\mathbf{j})$ m s^{-1}. When $t = 0$, the displacement of the particle relative to a fixed origin O is $(-\mathbf{i} + 3\mathbf{j})$ m. Find the displacement of the particle at time t.

Hint

Use $\mathbf{r} = \int \mathbf{v}\,dt$.

3. A particle is moving in a plane so that at time t seconds its acceleration $\mathbf{a}$ m s^{-2} is given by

$$\mathbf{a} = (1 - 6t)\mathbf{i} + 3\mathbf{j},\ t \geq 0$$

When $t = 0$ the particle has velocity $(\mathbf{i} + 2\mathbf{j})$ m s^{-1} and the position of the particle relative to a fixed origin O is $(4\mathbf{i} + \mathbf{j})$ m. Find:

a. the displacement of the particle from O at time t

b. the distance of the particle from O when $t = 3$.

Hint

Integrate $\mathbf{a}$ to find the velocity first, then integrate again to find the displacement.

4. E/P A particle is initially at rest at a fixed origin O. The particle moves in a plane such that its acceleration, $\mathbf{a}$ m s^{-2}, at time t seconds is given by $\cos \pi t\mathbf{i} + 2\sin \pi t\mathbf{j}$, $t \geq 0$. Find:

a. the velocity of the particle when $t = 3$ (5 marks)

b. the position vector of the particle when $t = 4$. (5 marks)

5. E/P At time t seconds a particle is moving in a plane with acceleration $(3t\mathbf{i} - 2\mathbf{j})$ m s^{-2}, where $t \geq 0$. When $t = 0$, the velocity of the particle is $2\mathbf{i}$ m s^{-1}. Find:

a. the velocity of the particle at time t seconds (3 marks)

b. the speed of the particle when $t = 2.4$ (3 marks)

c. the direction in which the particle is moving when $t = 4.8$. (3 marks)

6. E/P At time t seconds a particle, P is moving in a plane with velocity $\mathbf{v}$ m s^{-1}, where

$$\mathbf{v} = (4t + 1)\mathbf{i} - 2k\mathbf{j},\ t \geq 0$$

When $t = 0$, the displacement of the particle is $\mathbf{j}$ m.

a. Find an expression for the displacement of the particle. (3 marks)

A second particle, Q, is also moving in a plane. Its velocity is $(5\mathbf{i} - 3t\mathbf{j})$ m s^{-1}, where $t \geq 0$. When $t = 0$, the displacement of Q is $2\mathbf{j}$ m. Given that the particles collide, find:

b. the value of k (3 marks)

c. the position vector of the point of collision. (3 marks)

7. E/P A particle, P is moving in a plane with acceleration $\mathbf{a}$ m s^{-2}, where

$$\mathbf{a} = (3t - 1)\mathbf{i} + \mathrm{e}^t\mathbf{j},\ t \geq 0$$

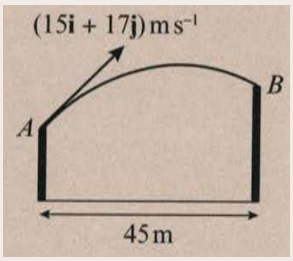
When $t = 0$, the velocity of the particle is $(7\mathbf{i} - 4\mathbf{j})$ m s^{-1}. Show that the particle never moves in a direction parallel to $\mathbf{j}$. (6 marks)

Problem solving: Set A

Practice Book, printed page 75.

Bronze

For this question take $g = 10$ m s^{-2}. A particle is projected with a velocity of $(15\mathbf{i} + 17\mathbf{j})$ m s^{-1} from a point A at the top of a building that stands on a horizontal plane. The particle moves freely under gravity until it lands at point B on top of a second building that stands on the same horizontal plane. The horizontal distance between the two buildings is 45 m. Find:

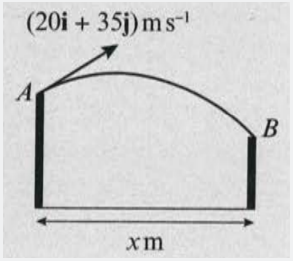


a. the velocity of the particle when it strikes B, in the form $(a\mathbf{i} + b\mathbf{j})$ m s^{-1} (4 marks)

b. the difference in height between the two buildings. (3 marks)

Silver

For this question, take $g = 10$ m s^{-2}. A particle is projected with a velocity of $(20\mathbf{i} + 35\mathbf{j})$ m s^{-1} from point A on the top of a building that stands on a horizontal plane. The particle moves freely under gravity until it lands at point B on top of a second building that stands on the same horizontal plane.



a. Find the time at which the particle is travelling parallel to the vector $2\mathbf{i} + \mathbf{j}$. (3 marks)

The particle strikes B at a speed of 30 m s^{-1}. Given that B is below A,

b. find the horizontal distance, x m, between the two buildings. (7 marks)

Gold

For this question take $g = 10$ m s^{-2}. Two particles P and Q are projected simultaneously from the top of a cliff with velocities $(30\mathbf{i} + 10\mathbf{j})$ m s^{-1} and $(20\mathbf{i} + 20\mathbf{j})$ m s^{-1} respectively. Find:

a. the time at which the two particles are instantaneously travelling in the same direction (5 marks)

b. the exact distance between them at this point. (3 marks)

Problem solving: Set B

Practice Book, printed page 76. In these questions you may assume that $\mathbf{i}$ and $\mathbf{j}$ are the unit vectors due east and north respectively.

Bronze

At time t seconds a particle P has position vector $\mathbf{r}$ m relative to a fixed origin O where $\mathbf{r} = 4t^2\mathbf{i} + 3t(t^2 - 1)\mathbf{j}$, $t \geq 0$. Work out:

a. the velocity of P at time t seconds (3 marks)

b. the speed of the particle when $t = 6$, correct to 3 significant figures (2 marks)

c. the exact time at which the particle is moving in a north-easterly direction. (3 marks)

Silver

At time t seconds, where $t \geq 0$, a particle P moves in a horizontal plane in such a way that its velocity $\mathbf{v}$ m s^{-1} is given by $\mathbf{v} = 3\mathbf{i} + (t^{-\frac{1}{2}} + 1)\mathbf{j}$. When $t = 0$, P is at a point A. When $t = 9$, P is at a point B.

a. Find the distance AB. (5 marks)

Given that the position vector of the point A relative to a fixed origin O is $-2\mathbf{j}$ m,

b. find the exact time at which the particle is due east of O. (3 marks)

Gold

At time t seconds a particle P has position vector $\mathbf{r}$ m relative to a fixed origin O where $\mathbf{r} = \cos \pi t\mathbf{i} + 3\sin^2 \pi t\mathbf{j}$, $t \geq 0$. Find, correct to 3 significant figures, the first time at which the particle is moving parallel to the vector $\mathbf{i} - \mathbf{j}$. (7 marks)

Selected exam question bank questions

Question 23

At time $t = 0$, two hikers, Athena and Jeff, have position vectors relative to a fixed origin of $\begin{pmatrix}12\\-6\end{pmatrix}$ km and $\begin{pmatrix}2\\16\end{pmatrix}$ km respectively.

Athena walks with constant velocity $\begin{pmatrix}-1\\4\end{pmatrix}$ km h^{-1}, Jeff with constant velocity $\begin{pmatrix}2\\-2\end{pmatrix}$ km h^{-1}.

Determine whether the two hikers will meet, justifying your answer fully. (4 marks)

Question 27

A particle P is moving in a plane. At time t seconds, it has a position vector $\mathbf{r}$ m relative to a fixed origin, where

$$\mathbf{r} = 3t(kt^2 + 1)\mathbf{i} + (5t^2 - 2)\mathbf{j},\ t \geq 0$$

a. Find an expression for the velocity of P at time t seconds. (2 marks)

Given that when $t = 2$, P has a speed of $5\sqrt{241}$ m s^{-1},

b. show that $k = 2$. (4 marks)

Question 30

A particle P moves along the x-axis in the positive direction.

At time t seconds, the velocity of P is v m s^{-1} and its acceleration is $\frac{2}{3}e^{-\frac{2}{5}t}$ m s^{-2}.

When $t = 0$, the velocity of P is 1 m s^{-1}.

a. Express v in terms of t. (4 marks)

b. Find the velocity of P when $t = 3$. (2 marks)

Question 33

A particle of mass 4 kg moves under the action of a constant force $(6\mathbf{i} + 2\mathbf{j})$ N.

At time $t = 0$ seconds, the particle has position vector $(4\mathbf{i} - 2\mathbf{j})$ m and is moving with velocity $\mathbf{u}$ m s^{-1}.

At time $t = 6$ seconds, the particle has position vector $(-3\mathbf{i} + 8\mathbf{j})$ m.

Find $\mathbf{u}$. (6 marks)

Question 39

A particle is projected with velocity $(23\mathbf{i} + 18\mathbf{j})$ m s^{-1} from a point O that lies on horizontal ground. The particle moves freely under gravity. Using $g = 9.8$ m s^{-2}, work out:

a. the greatest height reached by the particle (2 marks)

b. the horizontal distance travelled by the particle (3 marks)

c. the time at which the particle is moving parallel to the vector $2\mathbf{i} - \mathbf{j}$. (3 marks)

Question 44

At time t seconds a particle is moving in a plane with acceleration $\mathbf{a}$ m s^{-2}, where

$$\mathbf{a} = 3(t+2)\mathbf{i} + 2(t-1)\mathbf{j},\ t \geq 0$$

When $t = 0$, the velocity of the particle is $-2\mathbf{i} + 3\mathbf{j}$ m s^{-1}.

Find:

a. the speed of the particle when $t = 1.2$ (6 marks)

b. the time when the particle is moving parallel to the vector $\mathbf{i} + \mathbf{j}$. (4 marks)

Answers

8.1 Vectors in kinematics

1. $(9\mathbf{i} - 13\mathbf{j})$ m

2.
- **a.** $(-2\mathbf{i} + 3\mathbf{j})$ m
- **b.** 3 seconds

3.
- **a.** $(80\mathbf{i} + 80\mathbf{j})$ m
- **b.** 113 m (3 s.f.)

4. $\sqrt{145}$ m s^{-1} on a bearing of 085.2° (3 s.f.)

5. $(1.5\mathbf{i} - \mathbf{j})$ m s^{-2}

6.
- **a.** $a = -11$, $b = -18$
- **b.** 79.5 m (3 s.f.)

7.
- **a.** $\mathbf{r} = \begin{pmatrix} 1540 - 220t \\ 2500 - 400t \end{pmatrix}$ m
- **b.** 2020 m (3 s.f.)
- **c.** 7 seconds

8.
- **a.** $\sqrt{29}$ m s^{-1}
- **b.** $\begin{pmatrix} 2.4 \\ 2 \end{pmatrix}$ m s^{-2}
- **c.** $\begin{pmatrix} 54.2 \\ 14 \end{pmatrix}$ m

9.
- **a.** Equate $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$ for A and B to find the walkers have the same position vector at $t = 4$, so they meet.
- **b.** $\begin{pmatrix} 10 \\ 2 \end{pmatrix}$ km

10.
- **a.** $6\mathbf{i} - 8\mathbf{j}$
- **b.** 6 seconds

8.2 Vector methods with projectiles

1.
- **a.** $\begin{pmatrix} 15 \\ 5.4 \end{pmatrix}$ m s^{-1}
- **b.** 15.9 m s^{-1} (3 s.f.)

2.
- **a.** $\begin{pmatrix} 12t \\ 19t - 4.9t^2 \end{pmatrix}$ m
- **b.** $\begin{pmatrix} 6 + 12t \\ 6 + 19t - 4.9t^2 \end{pmatrix}$ m

3.
- **a.** $\begin{pmatrix} 23t \\ 15 + 13t - 4.9t^2 \end{pmatrix}$ m
- **b.**
  - **i.** 3.52 seconds (3 s.f.)
  - **ii.** 81.0 m (3 s.f.)

4. 2.5 m

5.
- **a.** $a = 24$, $b = 24.5$
- **b.** 30.6 m (3 s.f.)

6.
- **a.** 13.9 seconds (3 s.f.)
- **b.** 464 m (3 s.f.)

7.
- **a.** 30.3 m (3 s.f.)
- **b.** 13.3 m s^{-1} (3 s.f.)

8. $p = 5$, $q = 46.8$

8.3 Variable acceleration in one dimension

1.
a. $2(\pi + 2)$ m s^{-1}
b. 2 m s^{-2}

2.
a. $\left(\mathrm{e}^t + \frac{1}{3}t^3 - 1\right)$ m s^{-1}
b. $\left(\mathrm{e}^3 + \frac{11}{4}\right)$ m

3.
a. $2(1 + 4\ln 4)$ m s^{-1}
b. 2 m s^{-2}

4. $\left(-2\ln t + \frac{\cos(2\pi t)}{4\pi^2} + 2t - \frac{1}{4\pi^2} - 2\right)$ m

5.
a. $-\frac{1}{2}$
b. $\left(-\frac{1}{2}t\sin t - \frac{1}{2}\cos t + \frac{1}{2}\right)$ m
c. 1 m

6.
a. 2
b. $-\frac{8}{\mathrm{e}}$

7.
a. $v = \frac{t}{\sqrt{t^2+15}}$ m s^{-1}
b. $a = \frac{15}{(t^2+15)^{\frac{3}{2}}}$ which is always positive, so v is increasing for all values of t. Max speed $= \frac{7}{8}$ m s^{-1}

8.4 Differentiating vectors

1.
a. $\left(54\mathbf{i} - \frac{5}{6}\mathbf{j}\right)$ m s^{-1}
b. 6.5 m s^{-1} at an angle of 22.6° (3 s.f.) below the positive **i**-direction.

2.
a. $-5\mathbf{j}$ m s^{-2}
b. 101° (3 s.f.) below the positive **i**-direction.

3.
a. $\sqrt{85}$ m
b. $(2\mathbf{i} + 24\mathbf{j})$ m s^{-2}

4. $9\mathrm{e}^6\mathbf{i}$ m s^{-2}

5.
a. 0.5 seconds
b. 4.22 m s^{-1}

6.
a. 4 m s^{-1}
b. 1 second
c. $(-\mathbf{i} + 3\mathbf{j})$ m

7.
a. 13.6 m s^{-1} (3 s.f.)
b. $\left((48t^2 + 48t + 24)\mathrm{e}^{2t(t+1)}\mathbf{i} - \mathrm{e}^t\mathbf{j}\right)$ m s^{-1}

8.
a. $\mathbf{r} = (5t^2 + 5t)\mathbf{i} + (3 - t^2)\mathbf{j}$
$\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = (10t + 5)\mathbf{i} - 2t\mathbf{j}$
$\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} = 10\mathbf{i} - 2\mathbf{j}$
So P moves with constant acceleration.
b. $10\sqrt{26}$

8.5 Integrating vectors

1.
a. $\left(\left(\frac{5}{2}t^2 - 2\right)\mathbf{i} + 2(t-1)\mathbf{j}\right)$ m s^{-1}
b. 20.9 m s^{-1}

2. $\left((3t-1)\mathbf{i} + \left(3 - \frac{2}{3}t^3\right)\mathbf{j}\right)$ m

3.
a. $\left(\left(\frac{1}{2}t^2 - t^3 + t + 4\right)\mathbf{i} + \left(\frac{3}{2}t^2 + 2t + 1\right)\mathbf{j}\right)$ m
b. 25.7 m

4.
a. $\frac{4}{\pi}\mathbf{j}$ m s^{-1}
b. $\frac{8}{\pi}\mathbf{j}$ m

5.
a. $\left(\left(\frac{3}{2}t^2 + 2\right)\mathbf{i} - 2t\mathbf{j}\right)$ m s^{-1}
b. 11.7 m s^{-1} (3 s.f.)
c. 14.7° (3 s.f.)

6.
a. $(t(2t+1)\mathbf{i} + (1-2kt)\mathbf{j})$ m s^{-1}
b. $\frac{5}{4}$
c. $(10\mathbf{i} - 4\mathbf{j})$ m

7. $\mathbf{a} = (3t-1)\mathbf{i} + \mathrm{e}^t\mathbf{j}$
$\mathbf{v} = \left(\frac{3}{2}t^2 - t\right)\mathbf{i} + \mathrm{e}^t\mathbf{j} + \mathbf{c}$
At $t = 0$, $\mathbf{v} = 7\mathbf{i} - 4\mathbf{j}$
$\mathbf{v} = \left(\frac{3}{2}t^2 - t + 7\right)\mathbf{i} + (\mathrm{e}^t - 5)\mathbf{j}$
In order for $\mathbf{v}$ to be parallel to $\mathbf{j}$, $\frac{3}{2}t^2 - t + 7 = 0$ must have a real solution. Since $b^2 - 4ac < 0$ there are no real values of t that satisfy $\frac{3}{2}t^2 - t + 7 = 0$.

Problem solving: Set A

Bronze

a. $(15\mathbf{i} - 13\mathbf{j})$ m s^{-1}
b. 6 m

Silver

a. 2.5 seconds
b. 115 m (3 s.f.)

Gold

a. 4 seconds
b. $40\sqrt{2}$ m

Problem solving: Set B

Bronze

a. $(8t\mathbf{i} + (9t^2 - 3)\mathbf{j})$ m s^{-1}
b. 325 m s^{-1}
c. $\dfrac{4 + \sqrt{43}}{9}$ seconds

Silver

a. 30.9 m (3 s.f.)
b. $(4 - 2\sqrt{3})$ seconds

Gold 0.447 seconds (3 s.f.)

Selected exam question bank answers

Question 23 Equating **i**-direction position:

$12 - t = 2 + 2t \Rightarrow t = \frac{10}{3}$

Equation **j**-direction position:

$-6 + 4t = 16 - 2t \Rightarrow t = \frac{11}{3}$

So the hikers will not meet.

Question 27

a. $\mathbf{v} = ((9kt^2 + 3)\mathbf{i} + 10t\mathbf{j})$ m s^{-1}

b. At $t = 2$ s, $\mathbf{v} = (36k + 3)\mathbf{i} + 20\mathbf{j}$ then use $|\mathbf{v}| = 5\sqrt{241}$ to find k.

Question 30

a. $\left(\frac{8-5e^{-\frac{2}{5}t}}{3}\right)$ m s^{-1}

b. 2.16 m s^{-1} (3 s.f.)

Question 33 $\mathbf{u} = \left(-\frac{17}{3}\mathbf{i} + \frac{1}{6}\mathbf{j}\right)$ m s^{-1}

Question 39

a. $\frac{810}{49}$ m

b. $\frac{4140}{49}$ m

c. 3.01 s (3 s.f.)

Question 44

a. 7.64 m s^{-1} (3 s.f.)

b. 0.602 seconds (3 s.f.)